NEEDLEMAN WUNSCH ALGORITHM (GLOBAL)

- Needleman Wunsch algorithm:
  e.g., Align: ATCG & TCG

I. SCORING MATRIX

Match = +1

Mismatch = -1

Gap penalty = -2

G	-8			
C	-6			
T	-4			
A	-2			
	0	-2	-4	-6
		T	**C**	**G**

II. INITIALIZE & FILL THE MATRIX: (BASED ON GAP PENALTIES)

4	**G**	-8			
3	**C**	-6			
2	**T**	-4			
1	**A**	-2	-1		
0		0	-2	-4	-6
			T	**C**	**G**

Each box can obtain values from 3 places:

Horizontal/besides: gaps

Vertical/below: gaps

Diagonal: Match/Mismatch

Value of all 3 possibilities calculated
D = D value + match or mismatch score
C = C value + gap score
R = R value + gap score

Highest value considered and placed in the box:
e.g., D = 0 + (-1) = -1
C = -2 -2 = -4
R = -2 -2 = -4
Here, highest value is of diagonal i.e., -1 hence R1, C2 or i,j (1,2) will have value -1

G	-8			
C	-6			
T	-4			
A	-2	-1		
	0	-2	-4	-6
		T	**C**	**G**

G	-8	??		
C	-6		??	
T	-4			
A	-2	-1	-3	
	0	-2	-4	-6
		T	**C**	**G**

G	-8	-5	-2	1
C	-6	-3	0	-2
T	-4	-1	-2	-4
A	-2	-1	-3	-5
	0	-2	-4	-6
		T	**C**	**G**

Final Matrix

III. TRACE BACK:

1. Find the highest score: here it is +1
2. Highest value trace back to zero
3. Check adjacent 3 cells & check from where the value is obtained
4. If arrow is diagonal, insert sequence characters, if vertical or horizontal arrow is there then a gap is placed

G	-8	-5	-2	1
C	-6	-3	0	-2
T	-4	-1	-2	-4
A	-2	-1	-3	-5
	0	-2	-4	-6
		T	**C**	**G**

IV. ALIGNMENT:

A T C G
— T C G

We have also discussed the following sequence alignments using Needleman Wunsch algorithm in the lecture:

Nucleotide Sequence:

CTCGCAGC
CATTCAC

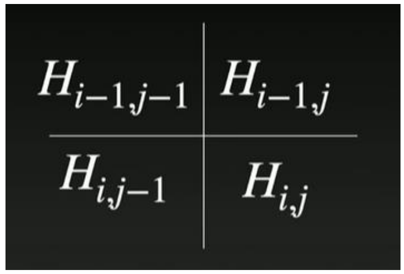
Protein Sequence:

FKHMEDPLE

FMDTPLNE

SMITH WATERMAN ALGORITHM (LOCAL)

- For Local alignment
- Most similar subsequence



➢ Smith Waterman Algorithm:

I. **Scoring matrix/Grid creation**

- Deciding the scores:
  Match = +1, Mismatch = -1, Gap penalty = -2

- Inserting gap:

- Initializing the matrix:
  Binary value insertion: If value is -ve then insert 0

- Filling the matrix:
  Value of all 3 possibilities calculated:
  D = D value + match or mismatch score
  C = C value + gap score
  R = R value + gap score

  If value is -ve then insert 0
  +ve values inserted as it is

II. **Traceback & align:**

- Start with the highest value
- Follow the arrows from where it is derived
- Follow till we reach 0
- Follow the same process until we get meaningful, non-overlapping regions

E.g.1

		T	G	T	C
	0	0	0	0	0
G	0	0	1	0	0
T	0	1	0	2	1
C	0	0	0	1	3
A	0	0	0	0	2

```
  G T C A
   | | |
T G T C
```

E.g. 2

Sequence 1: "A G C T A"

Sequence 2: "A C A C T A"

		A	G	C	T	A
	0	0	0	0	0	0
A	0	2←	0←	0←	0←	2←
C	0	0↑	1↖	2←	0←	0↑
A	0	2↖	0↑	1↖	1↖	2←
C	0	0↑	1↖	2↖	0↑	0↑
T	0	0↑	0↑	0↑	4↖	2←
A	0	2↖	0↑	0↑	2↑	6↖